

CSE 344 - Systems Programming

Homework #3

Implement Thread Synchronization using Mutexes and Semaphores

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Objective

The goal of this assignment was to simulate a ground station that coordinates communication between Earth and satellites. In this simulation, satellites request support from engineers for updates. A fixed number of engineers (3) are available to serve incoming satellite requests. Each satellite has a limited window (timeout) to receive support and is prioritized based on urgency.

Thread synchronization had to be managed using **POSIX threads (pthreads)**, **mutexes**, and **semaphores**.

Key Concepts Applied

- Thread synchronization using `pthread_mutex_t`
- Inter-thread signaling using `sem_t`
- Priority-based queueing
- Timeout handling via `sem_timedwait()`
- Shared resource protection
- Avoiding race conditions and memory errors

Program Flow Overview

Threads:

- Satellite Threads (`satellite`)
 - Request engineer support with a timeout (5).

- If served within the timeout, it prompts success.
- If not, it prompts a timeout and exits without service.
- Engineer Threads (engineer)
 - Wait for satellite requests (sem_wait).
 - Process highest-priority request from the queue.
 - Signal the satellite after processing (sem_post).

Synchronization:

- engineerMutex: Protects access to shared data: requestQueue, queueSize, and availableEngineers.
- newRequest (**semaphore**): Notifies engineers that a new satellite has arrived.
- requestHandled (**semaphore**): Used by engineers to notify satellites that their request has been served.

Important Structures

- Each satellite has a unique ID and a randomly assigned priority between 1–5.

```
typedef struct {
    int id;
    int priority;
} SatelliteReq;
```

Timeout Handling

Each satellite sets a timeout window (MAX_TIMEOUT = 5s). If no engineer becomes available during that time:

- The satellite removes itself from the queue.
- Prompts a timeout event.
- Exits without receiving service.

This prevents starvation and simulates real-world satellite communication limits.

Queue Management

A **priority queue** is implemented as an array of pointers (SatelliteReq* requestQueue[]) with insertion based on descending priority.

- enqueue() inserts a satellite into the queue based on priority.
- dequeue() removes and returns the highest-priority satellite.

All queue operations are protected by `engineerMutex` to ensure thread safety.

Execution Summary

- 3 engineer threads and 5 satellite threads are created.
- Satellites arrive at random intervals with random priorities.
- Engineers process requests in priority order.
- Timeout logic ensures satellites are not left waiting indefinitely.
- Proper cleanup (`pthread_join`, `sem_destroy`, `pthread_mutex_destroy`) is performed.

Compilation & Execution

```
$ make
```

```
$ make run
```

Example Scenarios

1. Engineers created with IDs: 1, 2, 3. 2. Satellite #1(2) queued. 3. E3 processing S1. 4. Satellite #2(5) queued. 5. E1 processing S2. 6. Satellite #3(1) queued. 7. E2 processing S3. 8. Satellite #4(3) queued. 9. E3 processed S1. 10. E3 processing S4. 11. S1 completed. 12. Satellite #5(1) queued. 13. S5 had timeout. 14. E3 completed S4. 15. S2 updated. 16. E1 completed S2. 17. E2 completed S3.	<pre> koixos@debian12:~/workspace/hw03\$ make run ./hw03 [SUCCESS] Created engineer with 1 [SUCCESS] Created engineer with 2 [SUCCESS] Created engineer with 3 [SUCCESS] Created satellite 1 with priority 2 [SATELLITE] Satellite 1 (Priority 2): Requesting engineer support [SATELLITE] Satellite 1 (Priority 2): Added to queue. [SYSTEM] Pending requests: 1 [ENGINEER] Engineer 3: Processing satellite 1 (Priority 2) [SYSTEM] Available engineers: 2 [SUCCESS] Created satellite 2 with priority 5 [SATELLITE] Satellite 2 (Priority 5): Requesting engineer support [SATELLITE] Satellite 2 (Priority 5): Added to queue. [SYSTEM] Pending requests: 1 [SUCCESS] Created satellite 3 with priority 1 [ENGINEER] Engineer 1: Processing satellite 2 (Priority 5) [SYSTEM] Available engineers: 1 [SUCCESS] Created satellite 4 with priority 3 [SATELLITE] Satellite 3 (Priority 1): Requesting engineer support [SATELLITE] Satellite 3 (Priority 1): Added to queue. [SYSTEM] Pending requests: 1 [ENGINEER] Engineer 2: Processing satellite 3 (Priority 1) [SYSTEM] Available engineers: 0 [SATELLITE] Satellite 4 (Priority 3): Requesting engineer support [SATELLITE] Satellite 4 (Priority 3): Added to queue. [SYSTEM] Pending requests: 1 [ENGINEER] Engineer 3: Completed satellite 1 update. [SYSTEM] Available engineers: 1 [ENGINEER] Engineer 3: Processing satellite 4 (Priority 3) [SYSTEM] Available engineers: 0 [SATELLITE] Satellite 1 (Priority 2): UPDATED - Leaving. [SUCCESS] Created satellite 5 with priority 1 [SATELLITE] Satellite 5 (Priority 1): Requesting engineer support [SATELLITE] Satellite 5 (Priority 1): Added to queue. [SYSTEM] Pending requests: 1 [SATELLITE] Satellite 5 (Priority 1): TIMEOUT - Leaving without update. [ENGINEER] Engineer 3: Completed satellite 4 update. [SYSTEM] Available engineers: 1 [SATELLITE] Satellite 2 (Priority 5): UPDATED - Leaving. [ENGINEER] Engineer 1: Completed satellite 2 update. [SYSTEM] Available engineers: 2 [ENGINEER] Engineer 2: Completed satellite 3 update. [SYSTEM] Available engineers: 3 [SUCCESS] All operations completed successfully. koixos@debian12:~/workspace/hw03\$ █ </pre>
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1. Engineers created with IDs: 1, 2, 3.	gcc -Wall -Wextra -o hw03 200104004039.c -lpthread koixos@debian12:~/workspace/hw03\$ make run
2. Satellite #1(4) queued.	./hw03
3. E2 processing S1.	[SUCCESS] Created engineer with 1
4. Satellite #2(1) queued.	[SUCCESS] Created engineer with 2
5. E1 processing S2.	[SUCCESS] Created engineer with 3
6. E2 processed S1.	[SUCCESS] Created satellite 1 with priority 4
7. S1 completed.	[SATELLITE] Satellite 1 (Priority 4): Requesting engineer support
8. Satellite #3(1) queued.	[SATELLITE] Satellite 1 (Priority 4): Added to queue.
9. E3 processing S3.	[SYSTEM] Pending requests: 1
10. Satellite #4(1) queued.	[ENGINEER] Engineer 2: Processing satellite 1 (Priority 4)
11. E2 processing S4.	[SYSTEM] Available engineers: 2
12. E1 completed S2.	[SUCCESS] Created satellite 2 with priority 1
13. Satellite #5(3) queued.	[SATELLITE] Satellite 2 (Priority 1): Requesting engineer support
14. E1 processing S5.	[SATELLITE] Satellite 2 (Priority 1): Added to queue.
15. S5 updated.	[SYSTEM] Pending requests: 1
16. E3 completed S3.	[ENGINEER] Engineer 1: Processing satellite 2 (Priority 1)
17. S2 updated.	[SYSTEM] Available engineers: 1
18. E1 completed S5.	[ENGINEER] Engineer 2: Completed satellite 1 update.
19. E2 completed S4.	[SYSTEM] Available engineers: 2
	[SATELLITE] Satellite 1 (Priority 4): UPDATED - Leaving.
	[SUCCESS] Created satellite 3 with priority 1
	[SATELLITE] Satellite 3 (Priority 1): Requesting engineer support
	[SATELLITE] Satellite 3 (Priority 1): Added to queue.
	[SYSTEM] Pending requests: 1
	[ENGINEER] Engineer 3: Processing satellite 3 (Priority 1)
	[SYSTEM] Available engineers: 1
	[SUCCESS] Created satellite 4 with priority 1
	[SATELLITE] Satellite 4 (Priority 1): Requesting engineer support
	[SATELLITE] Satellite 4 (Priority 1): Added to queue.
	[SYSTEM] Pending requests: 1
	[ENGINEER] Engineer 2: Processing satellite 4 (Priority 1)
	[SYSTEM] Available engineers: 0
	[ENGINEER] Engineer 1: Completed satellite 2 update.
	[SYSTEM] Available engineers: 1
	[SUCCESS] Created satellite 5 with priority 3
	[SATELLITE] Satellite 5 (Priority 3): Requesting engineer support
	[SATELLITE] Satellite 5 (Priority 3): Added to queue.
	[SYSTEM] Pending requests: 1
	[ENGINEER] Engineer 1: Processing satellite 5 (Priority 3)
	[SYSTEM] Available engineers: 0
	[SATELLITE] Satellite 5 (Priority 3): UPDATED - Leaving.
	[ENGINEER] Engineer 3: Completed satellite 3 update.
	[SYSTEM] Available engineers: 1
	[SATELLITE] Satellite 2 (Priority 1): UPDATED - Leaving.
	[ENGINEER] Engineer 1: Completed satellite 5 update.
	[SYSTEM] Available engineers: 2
	[ENGINEER] Engineer 2: Completed satellite 4 update.
	[SYSTEM] Available engineers: 3
	[SUCCESS] All operations completed successfully.
	koixos@debian12:~/workspace/hw03\$ █